PROJECT PROPOSAL

MEDISHIELD AI – SMART MEDICINE SAFETY AND COUNTERFEIT DETECTION PLATFORM

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Project Type: Data Analysis, Medicine Verification and Interactive Dashboard Development

1. Introduction

Counterfeit medicines are a major concern in the healthcare and pharmaceutical industries. These medicines may contain incorrect ingredients, improper dosages, or misleading manufacturer information, potentially affecting patient safety. Identifying suspicious medicines and analyzing pharmaceutical data are important for improving medicine safety and transparency.

MediShield AI is a data analytics and visualization project designed to analyze medicine information, verify manufacturer details, identify suspicious records, and visualize medicine verification trends. The system uses structured pharmaceutical datasets, Python-based data processing, and Power BI dashboards to provide meaningful insights into medicine verification activities.

2. Problem Statement

The increasing circulation of counterfeit and unauthorized medicines creates challenges for patients, pharmacies, healthcare providers, and regulatory authorities. Traditional verification methods may involve manual inspection, which can be time-consuming and prone to errors.

There is a need for a data-driven system that can compare medicine information against a reference database, identify manufacturer mismatches, classify suspicious records, and present verification statistics through interactive visualizations.

3. Objectives

The main objectives of the proposed system are:

1. To develop a structured dataset containing medicine, manufacturer, prescription, and verification information.
2. To clean and preprocess pharmaceutical data using Python.
3. To compare scanned medicine manufacturer details with reference manufacturer information.
4. To identify suspicious medicine records based on verification mismatches.
5. To calculate risk scores for medicine verification records.
6. To analyze medicine categories, manufacturers, prescriptions, and verification trends.
7. To develop an interactive Power BI dashboard for visualizing analytical results.
8. To support data-driven identification of potential counterfeit medicine risks.

4. Proposed System

The proposed MediShield AI system performs medicine verification and pharmaceutical data analysis using a structured reference database.

The system maintains medicine information, including medicine name, generic name, category, dosage, dosage form, and manufacturer details.

During verification, the scanned medicine information is compared with the corresponding reference record. If the manufacturer details match, the record is classified as verified. If a mismatch is detected, the record is flagged as suspicious for further investigation.

A risk score is assigned to support prioritization of suspicious records. Python is used for data cleaning, validation, and analysis, while Power BI is used to present the results through interactive dashboards.

The system is intended as an analytical prototype and does not independently establish the authenticity or pharmaceutical quality of a medicine.

5. Dataset Description

The project uses five interconnected datasets:

Dataset 1: Prescription Data

Contains prescription details such as prescription ID, patient information, medicine name, dosage, frequency, duration, doctor name, and pharmacy name.

Dataset 2: Medicine Database

Contains reference information about medicines, including medicine ID, medicine name, generic name, category, dosage, dosage form, and manufacturer name.

Dataset 3: Manufacturer Data

Contains manufacturer details, including manufacturer ID, manufacturer name, number of medicines, verified medicines, suspicious medicines, and location.

Dataset 4: Dashboard Summary

Contains summarized analytical information such as total medicines, total scans, verified records, suspicious records, verification percentage, category distribution, and monthly scan counts.

Dataset 5: Scans Log

Contains medicine verification records, including scan ID, scan date, medicine ID, scanned manufacturer, reference manufacturer, batch number, QR code, expiry date, match result, risk score, and location.

6. Methodology

The project follows a structured data analytics workflow.

Step 1: Data Collection

Create and organize pharmaceutical datasets containing medicine details, manufacturer information, prescriptions, and verification records.

Step 2: Data Cleaning

Use Python and Pandas to identify missing values, remove duplicate records, standardize data formats, and correct inconsistencies.

Step 3: Data Preprocessing

Prepare cleaned datasets by validating medicine IDs, checking manufacturer references, formatting dates, and organizing records for analysis.

Step 4: Medicine Verification

Compare scanned manufacturer information against the reference medicine database to identify matching and mismatching records.

Step 5: Data Analysis

Analyze verification results, medicine categories, manufacturer distributions, risk scores, and monthly verification trends using Python.

Step 6: Data Visualization

Develop an interactive Power BI dashboard containing KPI cards, donut charts, bar charts, line charts, geographic visualizations, and detailed verification tables.

Step 7: Result Interpretation

Interpret the analytical results to identify suspicious records, observe verification patterns, and summarize medicine safety indicators.

7. Technologies Used

Technology	Purpose
Python	Data processing and analysis
Pandas	Data cleaning and manipulation
NumPy	Numerical operations
Matplotlib / Seaborn	Python-based data visualization
Microsoft Excel	Dataset creation and storage

Power BI	Interactive dashboard development
Visual Studio Code	Python development
GitHub	Project repository and version control

8. Expected Outcomes

The expected outcomes of MediShield AI include:

1. A structured pharmaceutical dataset containing medicine and manufacturer information.
2. Cleaned and validated datasets suitable for analysis.
3. Identification of suspicious manufacturer mismatches.
4. Risk scores for prioritizing verification records.
5. Statistical analysis of medicine verification activities.
6. Interactive dashboards displaying verification trends and medicine distributions.
7. Improved visibility into potential counterfeit medicine risks.
8. A reusable data analytics workflow for pharmaceutical verification records.

9. Applications

MediShield AI can be used as a prototype for pharmaceutical data analysis, medicine verification research, pharmacy inventory monitoring, manufacturer information validation, and healthcare analytics.

The analytical approach may also support future medicine traceability systems and regulatory monitoring tools when integrated with authenticated pharmaceutical data sources.

10. Future Scope

The project can be extended by integrating real-time QR code scanning, optical character recognition (OCR), authenticated manufacturer databases, batch-level verification, and machine learning models for advanced anomaly detection.

Future enhancements may include automated alerts, mobile applications, real-time dashboards, and secure integration with pharmaceutical supply-chain systems.

11. Conclusion

MediShield AI – Smart Medicine Safety and Counterfeit Detection Platform provides a data-driven approach to analyzing pharmaceutical information and identifying potentially suspicious medicine records.

By combining Python-based data cleaning, manufacturer verification, statistical analysis, and Power BI visualization, the project transforms structured medicine data into meaningful analytical insights.

The proposed system demonstrates how data analytics and visualization techniques can support medicine safety monitoring, improve transparency, and assist in identifying records that require further verification.